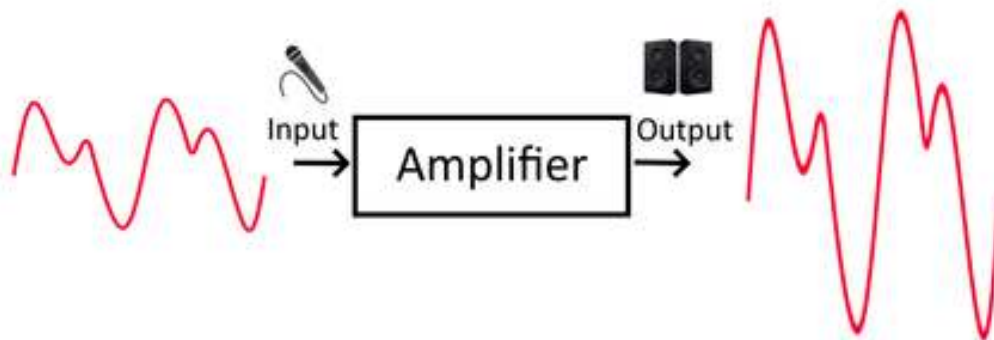




Egypt-Japan University of Science and Technology
Faculty of Engineering
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Project Report

Audio Amplifier Using LM386 IC



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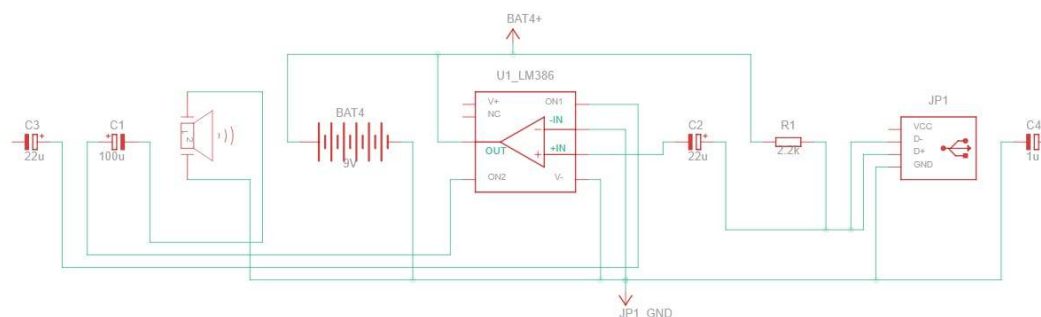
Course: ECE 212 – Electronics Engineering Lab

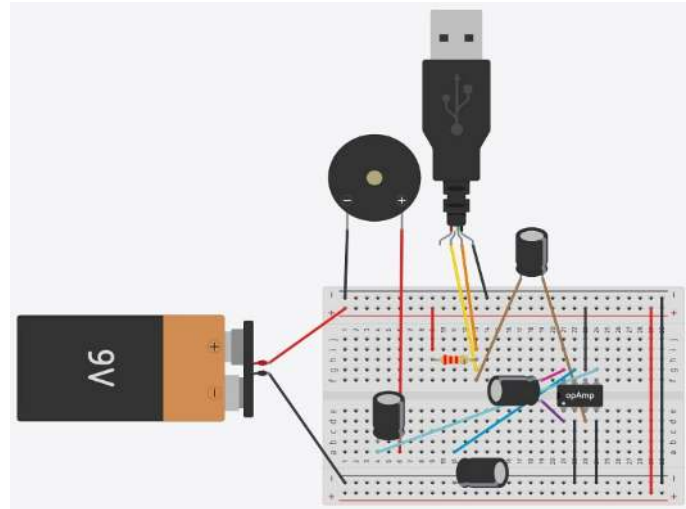
Aim of the Project

To build a simple audio amplifier that takes a weak audio signal from an AUX source (phone, laptop, etc.) and plays it through a small speaker with clear, amplified sound. This allows us to listen to music or speech from portable devices without needing powered external speakers. The project demonstrates how a low-cost integrated circuit (the LM386) can boost tiny electrical signals to a level that drives a speaker directly. By completing this project, we also learn the practical roles of resistors, capacitors, and proper grounding in a real audio circuit.

Components Used

Component	Value	Quantity
LM386 Op-Amp	—	1
Resistor	2.2 k Ω	1
Capacitor (electrolytic)	22 μ F	2
Capacitor (electrolytic)	100 μ F	1
Capacitor (electrolytic)	1 μ F	1
Speaker	8 Ω , 0.25 W	1
AUX cable	—	1
Jumper wires	—	As needed
Breadboard	—	1
Power supply	9V battery	1





Circuit Connections

- **Pin 1 to Pin 8** → 22 μF capacitor
- **Pin 2** → Ground
- **Pin 3** → One end of the second 22 μF capacitor (the other end of this capacitor goes to the junction of the AUX signal and the 2.2 k Ω resistor)
- **Pin 4** → Ground
- **Pin 5** → Positive side of 100 μF capacitor → then to speaker (+)
- **Pin 6** → +9V from battery
- **Pin 7** → 1 μF capacitor → Ground
- **AUX cable:**
 - Signal wire (left and right channel) → connected to one end of the **2.2 k Ω resistor** and to the **22 μF input capacitor** (which then goes to pin 3)
 - Ground wire of AUX → connected to circuit ground
- **Speaker** negative terminal → Ground

How the Circuit Works

Power Supply

The 9V battery powers pin 6. Pin 4 is grounded. The internal circuitry of the LM386 amplifies the audio signal using this power.

Audio Input (Pin 3)

- The AUX signal (AC + possible DC offset) comes from your phone.
- It passes through a **22 μ F coupling capacitor** that blocks DC and only lets the AC audio through.
- Then it goes to pin 3.
- The **2.2 k Ω resistor** connects from the AUX signal line to ground. This prevents the input from floating and reduces noise pickup.

Gain Control (Pin 1 & Pin 8)

- Without a capacitor between pin 1 and pin 8, the internal gain is 20.
- By adding a **22 μ F capacitor**, the gain increases to **200** – much louder volume.

Amplification Inside LM386

The IC uses internal transistors to boost the weak audio signal. It is specially designed to drive a small speaker directly.

Output to Speaker (Pin 5)

- The amplified signal leaves pin 5.
- A **100 μ F capacitor** passes only the AC audio to the speaker and blocks any DC voltage (which would overheat the speaker coil and damage it).
- The **8 Ω speaker** converts the electrical signal into sound.

Noise Reduction (Pin 7)

- Pin 7 is the internal “bypass” pin.
- A **1 μ F capacitor** from pin 7 to ground filters out internal power supply noise, giving cleaner sound.

Calculations

1. Voltage Gain of the Amplifier

The LM386 normally has a default voltage gain of: Gain = 20

When a capacitor is connected between pin 1 and pin 8, the gain increases.

In this project, a 22 μ F capacitor is connected between pin 1 and pin 8, therefore:

Gain = 200

This means the amplifier attempts to amplify the input signal by a factor of 200, limited by the 9 V power supply and the LM386 internal circuitry.

Example:

If the input signal amplitude is: $V_{in} = 10 \text{ mV} = 0.01 \text{ V}$

Then: $V_{out} = \text{Gain} \times V_{in} = 200 \times 0.01 = 2 \text{ V}$

Therefore, a very small audio signal can produce a much larger output signal capable of driving the speaker.

2. Output Power of the Circuit

The speaker used in this project has the following rating:

- Speaker resistance = 8Ω
- Maximum speaker power = 0.25 W

Therefore, the speaker should not receive more than 0.25 W continuously to avoid possible damage.

The RMS voltage corresponding to 0.25 W across an 8Ω speaker can be calculated using: $P = (V_{rms}^2) \div R$

Rearranging the equation: $V_{rms} = \sqrt{(P \times R)}$

Substitute the values: $V_{rms} = \sqrt{(0.25 \times 8)} = \sqrt{2} \approx 1.41 \text{ V}$

Therefore:

- RMS output voltage $\approx 1.41 \text{ V}$
- Safe speaker power $\approx 0.25 \text{ W}$

This output level is sufficient for a small portable audio amplifier and produces clear audible sound without overloading the speaker.

3. Input Impedance

The input impedance is mainly determined by the resistor connected between the input signal and ground.

Given: $R = 2.2 \text{ k}\Omega$

Therefore: Input impedance $\approx 2.2 \text{ k}\Omega$

This value is safe for phones, laptops, and AUX audio outputs.

4. Bypass Capacitor on Pin 7

A $1 \mu\text{F}$ capacitor is connected between pin 7 and ground.

Purpose of this capacitor:

- reduces internal noise
- improves sound quality
- stabilizes the amplifier operation

This results in cleaner and more stable audio output.

5. Power Supply Requirement

The circuit operates using: $V_{cc} = 9\text{ V}$

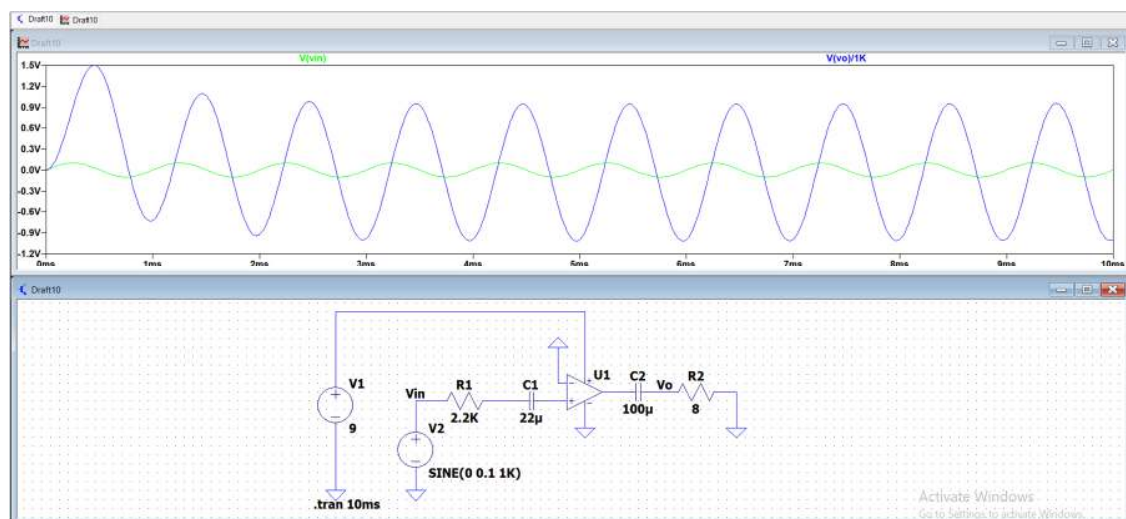
The LM386 typically operates between: 4 V and 12 V

A 9 V battery is suitable because it provides:

- stable operation
- sufficient output volume
- portable power source

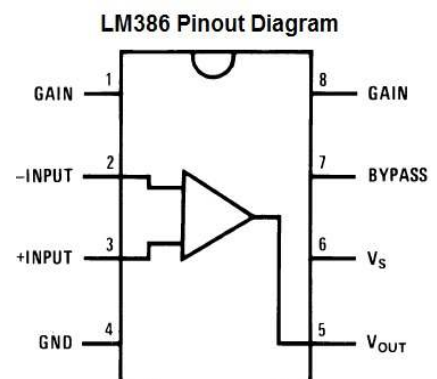
without requiring an external laboratory power supply.

LTspice Simulation Results



LTspice. The schematic included:

- LM386 model (Op-Amp)
- Input: 1 kHz sine wave with small amplitude
- 2.2 kΩ input resistor, 22 μF input capacitor
- 22 μF gain capacitor between pins 1 and 8
- 100 μF output capacitor and 8 Ω speaker load
- 1 μF bypass capacitor on pin 7



- 9V DC power supply

Simulation settings:

- **.tran 10ms** – transient analysis for 10 milliseconds (10 cycles of 1 kHz)
- Plotted: V(vin) = input voltage, and V(out) = output voltage across the speaker.

Observed Waveforms

The simulation plot displays:

- **Input signal (V(vin))**: A clean sine wave with amplitude ~ 0.1 V (100 mV peak).
- **Output signal (V(out))**: A larger sine wave with the 22 μ F gain capacitor, the theoretical voltage gain is approximately 200.

The plot shows an amplified sine wave without distortion, confirming the LM386 works correctly.

What the Simulation Confirms

- The **gain of 200** is achieved (loud volume).
- No visible distortion or clipping for moderate input levels.
- The output coupling capacitor (100 μ F) successfully blocks DC – the output waveform is centered around 0 V.
- The input coupling capacitor (22 μ F) blocks any DC offset from the source.

The simulation matches the real behavior of your breadboard circuit.

Conclusion

The LM386 audio amplifier works correctly and efficiently. The gain is set to 200 for strong output, and all capacitors perform their roles: blocking DC, reducing noise, and passing clean audio. The LTspice simulation confirms the expected gain and signal quality. The circuit is suitable for small portable speakers, DIY projects, and learning about audio amplification. The calculated cutoff frequencies show that the amplifier passes most audible frequencies well, though low bass is slightly reduced to protect the small speaker. Future improvements like adding a potentiometer for volume control or a larger filter capacitor on the power supply would make the design even better for everyday use.